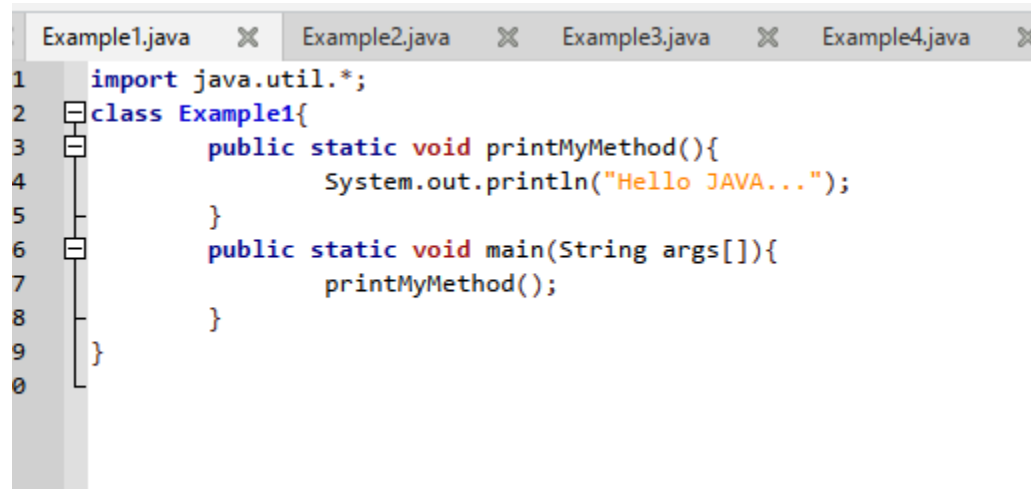
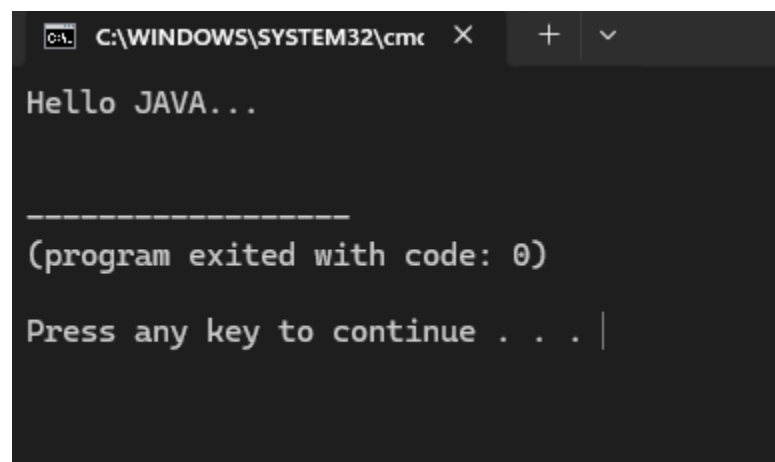


Q1.



```
Example1.java X Example2.java X Example3.java X Example4.java X
1  import java.util.*;
2  class Example1{
3      public static void printMyMethod(){
4          System.out.println("Hello JAVA...");
5      }
6      public static void main(String args[]){
7          printMyMethod();
8      }
9  }
10
```

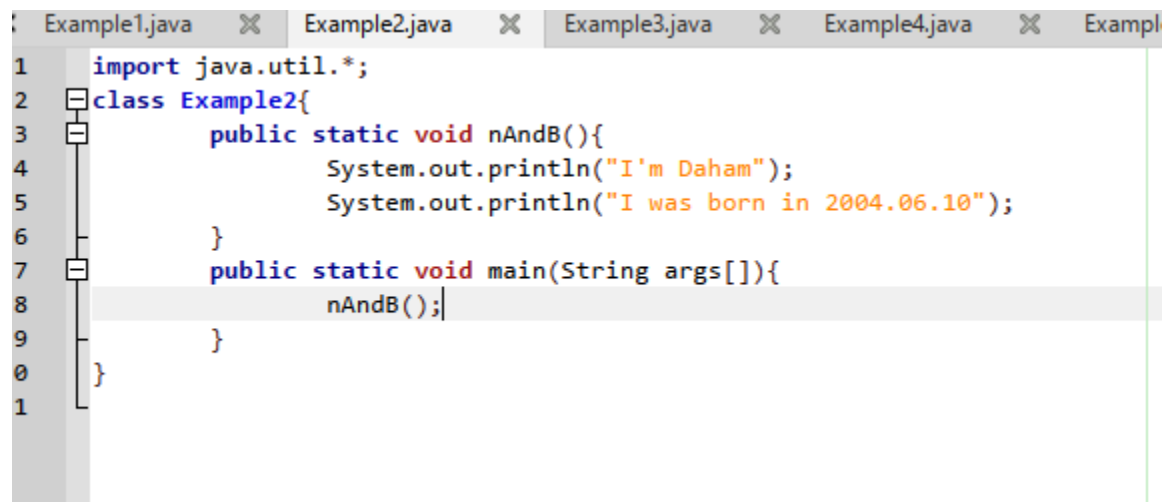
Output.



```
C:\WINDOWS\SYSTEM32\cmd X + v
Hello JAVA...

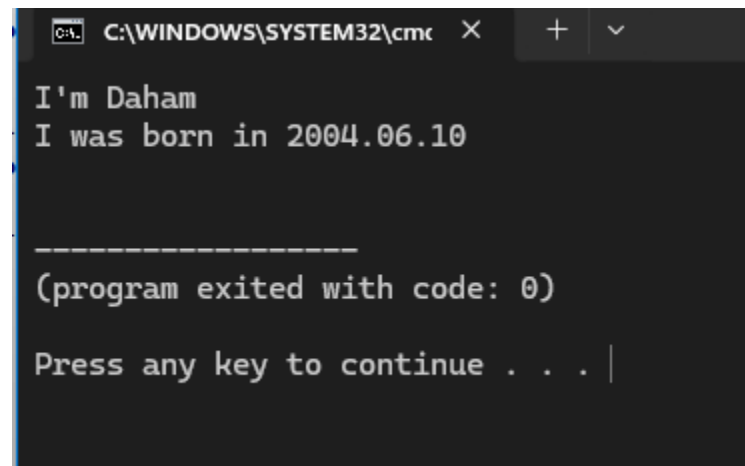
-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q2.



```
1 import java.util.*;
2 class Example2{
3     public static void nAndB(){
4         System.out.println("I'm Daham");
5         System.out.println("I was born in 2004.06.10");
6     }
7     public static void main(String args[]){
8         nAndB();
9     }
10 }
11 }
```

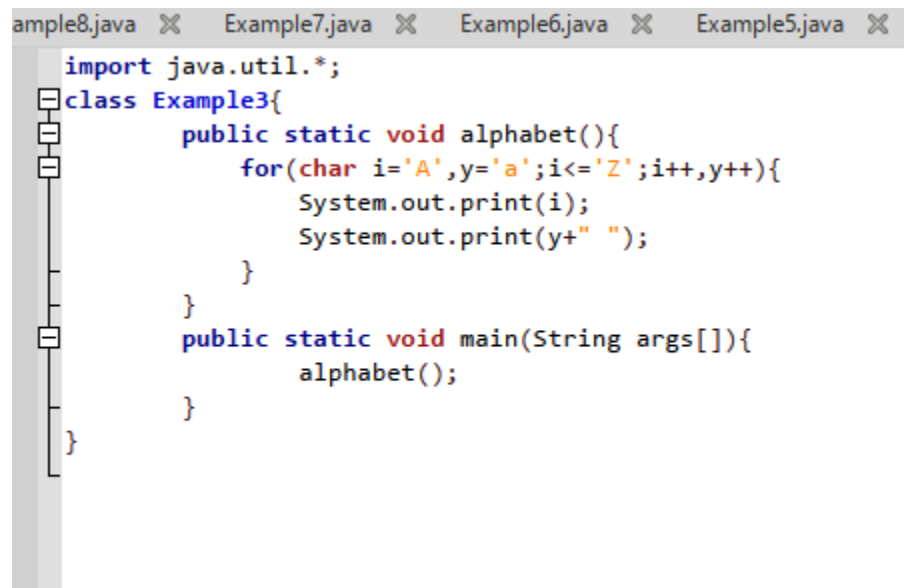
Output.



```
C:\WINDOWS\SYSTEM32\cmd
I'm Daham
I was born in 2004.06.10

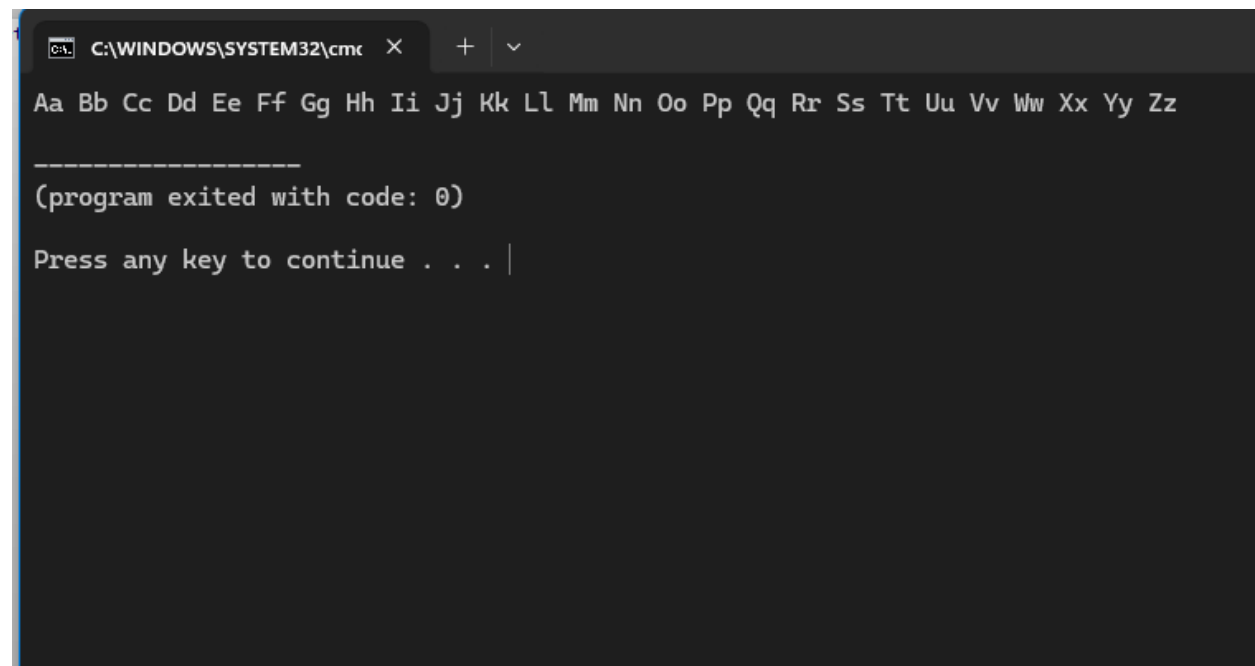
-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q3.



```
ample8.java ✕ Example7.java ✕ Example6.java ✕ Example5.java ✕  
import java.util.*;  
class Example3{  
    public static void alphabet(){  
        for(char i='A',y='a';i<='Z';i++,y++){  
            System.out.print(i);  
            System.out.print(y+" ");  
        }  
    }  
    public static void main(String args[]){  
        alphabet();  
    }  
}
```

Output.



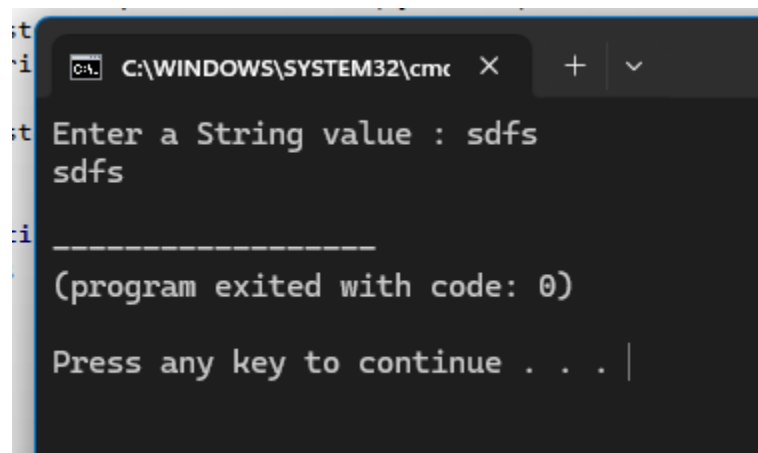
```
C:\WINDOWS\SYSTEM32\cmd X + v  
Aa Bb Cc Dd Ee Ff Gg Hh Ii Jj Kk Ll Mm Nn Oo Pp Qq Rr Ss Tt Uu Vv Ww Xx Yy Zz  
-----  
(program exited with code: 0)  
Press any key to continue . . . |
```

Q4.

```
import java.util.*;
class Example4{
    public static void scan(){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter a String value : ");
        String stri =input.nextLine();

        System.out.print(stri);
    }
    public static void main(String args[]){
        scan();
    }
}
```

Output.



```
C:\WINDOWS\SYSTEM32\cmd.exe
Enter a String value : sdfs
sdfs
-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q5.

```
Example8.java ✕ Example7.java ✕ Example6.java ✕ Example5.java ✕
import java.util.*;
class Example5{
    public static void leapY(){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter the year : ");
        int year = input.nextInt();

        if((year %4 == 0 && year % 100!=0) || (year%400==0)){
            System.out.print(year + " is a leap year");
        }else{
            System.out.print(year + " is not a leap year");
        }

    }

    public static void main(String args[]){
        leapY();
    }
}
```

Output.

```
Enter the year : 2044
2044 is a leap year

-----
(program exited with code: 0)

Press any key to continue . . . |
```

Q6.

```
import java.util.*;
class Example6{
    public static void calIn(){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter the num 1 : ");
        int num1 = input.nextInt();

        System.out.print("Enter the num 2 : ");
        int num2 = input.nextInt();

        if(num1>num2){
            System.out.print("--> "+num1+" > "+num2);
        }else if(num2>num1){
            System.out.print("--> "+num2+" > "+num1);
        }else{
            System.out.print("--> "+num2+" = "+num1);
        }
    }
    public static void main(String args[]){
        calIn();
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd
Enter the num 1 : 23
Enter the num 2 : 65
--> 65 > 23

-----
(program exited with code: 0)
Press any key to continue . . . |
```

07. **Program 01**

```
import java.util.*;
class Demo{
    public static void printName(){
        Scanner input=new Scanner(System.in);
        System.out.print("Input first name : ");
        String name1=input.nextLine();
        System.out.print("Input second name : ");
        String name2=input.nextLine();
        System.out.println(name1+" "+name2);
    }

    public static void main(String[] args){
        printName(); // line X
    }
}
```

Program 02

```
import java.util.*;
class Demo{
    public static void printName(String name1, String name2){
        System.out.println(name1+" "+name2);
    }

    public static void main(String[] args){
        Scanner input=new Scanner(System.in);
        System.out.print("Input first name : ");
        String name1=input.nextLine();
        System.out.print("Input second name : ");
        String name2=input.nextLine();

        printName(name1,name2); // Line Y
    }
}
```

I. What are the different methods invoking between Program 1 and Program 2?

Answer-

In first program it program is running in printName() method, and it calls in main() method.

But in second program some part of program run in main method and another part runs in printName() method, so that 2nd one has parameters and it also calls in main() method.

II. Can replace line Y in Program 2 with line X in Program 1 ? Give the reason for your answer.

Answer –

No,

Because of running Scanner in main method in program 2, we have to link methods by parameters and arguments ,Without arguments in main method, we cant run printName() method.

Q8.

```
imple8.java X
import java.util.*;
class Example8{
    public static void getWord(){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter the first word : ");
        char f = input.next().toUpperCase().charAt(0);

        System.out.print("Enter the Second word : ");
        char s = input.next().toUpperCase().charAt(0);

        if(f==s){
            System.out.print(true);
        }else{
            System.out.print(false);
        }
    }
    public static void main(String args[]){
        getWord();
    }
}
```

Outputs.

```
C:\WINDOWS\SYSTEM32\cmd X + v
Enter the first word : fwfw
Enter the Second word : frerh
true

-----
(program exited with code: 0)
Press any key to continue . . .
```

```
C:\WINDOWS\SYSTEM32\cmd X + v
Enter the first word : rrghrj
Enter the Second word : sghr
false

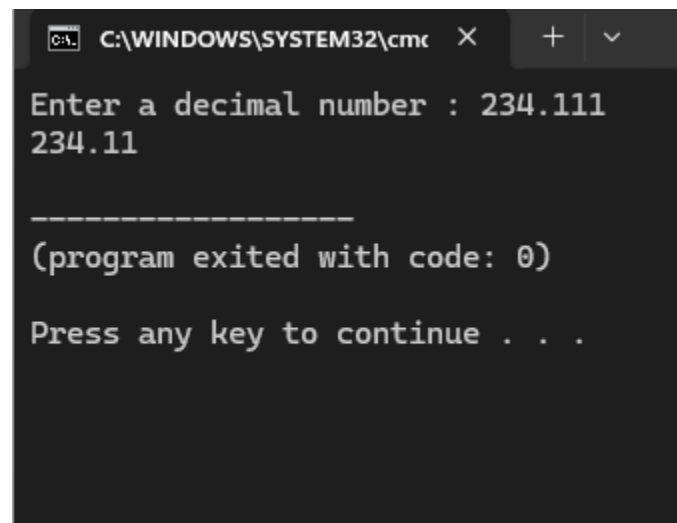
-----
(program exited with code: 0)
Press any key to continue . . .
```

Q9.

```
import java.util.*;
class Example9{
    public static void getNum(double num){
        System.out.printf("%.2f",num);
    }
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter a decimal number : ");
        double num = input.nextDouble();

        getNum(num);
    }
}
```

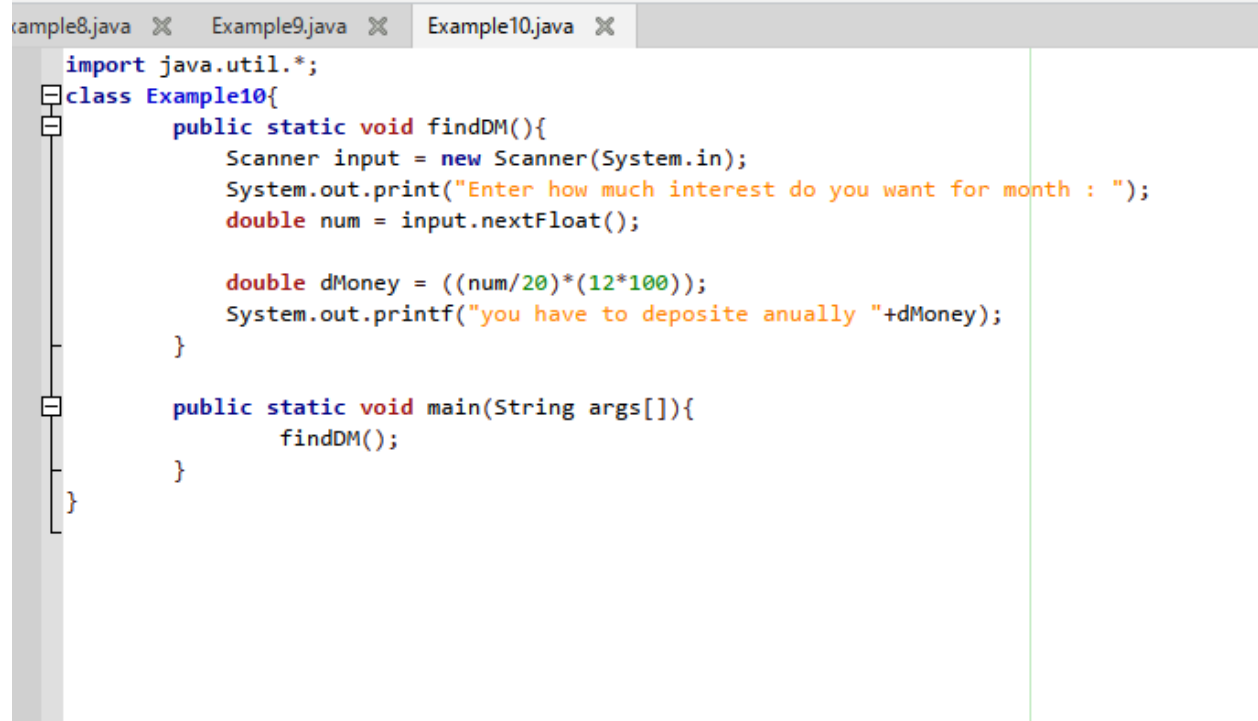
Output.



```
C:\WINDOWS\SYSTEM32\cmd.exe
Enter a decimal number : 234.111
234.11

-----
(program exited with code: 0)
Press any key to continue . . .
```

Q10.



```
import java.util.*;

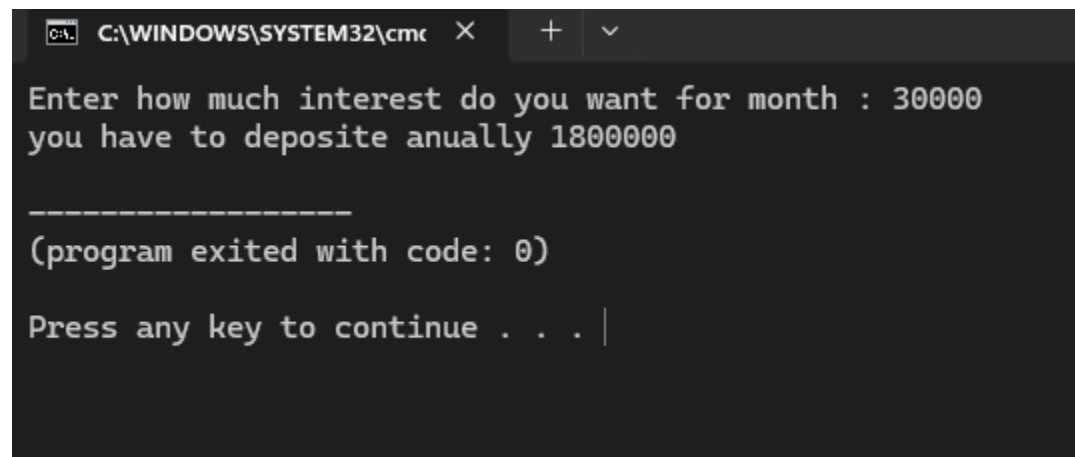
class Example10{

    public static void findDM(){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter how much interest do you want for month : ");
        double num = input.nextFloat();

        double dMoney = ((num/20)*(12*100));
        System.out.printf("you have to deposit anually "+dMoney);
    }

    public static void main(String args[]){
        findDM();
    }
}
```

Output.



```
C:\WINDOWS\SYSTEM32\cmd.exe X + v

Enter how much interest do you want for month : 30000
you have to deposit anually 1800000

-----
(program exited with code: 0)

Press any key to continue . . . |
```

Q11.

```
Example11.java  X  Example12.java  X  Example13.java  X  Example14.java  X  Exa
import java.util.*;
class Example11{
    public static void celconverter(double c){
        double f = (c*9/5)+32;
        double k = c+ 273.15;

        System.out.println("fahrenheit - "+f);
        System.out.print("Celsius - "+k);

    }
    public static void main(String args[]){
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter celsios value : ");
        double c = sc.nextDouble();

        celconverter(c);

    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd  X  +  v
Enter celsios value : 45
fahrenheit - 113.0
Celsius - 318.15
-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q12.

```
import java.util.*;
class Example12{
    public static void getPAndA(double an,double r){
        double area = 3.14 * r * r;
        double perimeter = 2 * 3.14 * r;
        System.out.printf("area - %.2f\n",area);
        System.out.printf("perimeter - %.2f",perimeter);
    }

    public static void main(String args[]){
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the angle : ");
        double an = sc.nextDouble();

        System.out.print("Enter the radius : ");
        double r = sc.nextDouble();

        getPAndA(an,r);
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd
Enter the angle : 34
Enter the radius : 4
area - 50.24
perimeter - 25.12

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q14.

```
import java.util.*;
class Example14{
    public static void add(int num1,int num2){
        System.out.print(num1+" + "+num2+" = "+(num1+num2));
    }
    public static void subtraction(int num1,int num2){
        System.out.print(num1+" - "+num2+" = "+(num1-num2));
    }
    public static void multiplication(int num1,int num2){
        System.out.print(num1+" * "+num2+" = "+(num1*num2));
    }
    public static void dividend(int num1,int num2){
        System.out.print(num1+" / "+num2+" = "+((double)num1/((double)num2));
    }
    public static void reminder(int num1,int num2) {
        System.out.print(num1+" % "+num2+" = "+(num1%num2));
    }
    public static void power(int num1,int num2){
        System.out.print(num1+" ^ "+num2+" = "+((int)Math.pow(num1,num2)));
    }

    public static void main(String args[]){
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the first number : ");
        int num1 = sc.nextInt();
        sc.nextLine();

        System.out.print("Enter the second number : ");
        int num2 = sc.nextInt();
        sc.nextLine();

        System.out.print("Enter the operator (+,-,*,/,%,^) : ");
        String op = sc.nextLine();

        switch(op){
            case "+":
                add(num1,num2);
                break;
            case "-":
                subtraction(num1,num2);
                break;
            case "*":
                multiplication(num1,num2);
                break;
            case "/":
                dividend(num1,num2);
                break;
            case "%":
                reminder(num1,num2);
                break;
            case "^":
                power(num1,num2);
                break;
        }
    }
}
```

Outputs.

```
C:\WINDOWS\SYSTEM32\cmd X + v
Enter the first number : 23
Enter the second number : 4
Enter the operator (+,-,*,/,%,^): ^
23 ^ 4 = 279841

-----
(program exited with code: 0)

Press any key to continue . . . |
```

```
C:\WINDOWS\SYSTEM32\cmd X + v
Enter the first number : 43
Enter the second number : 56
Enter the operator (+,-,*,/,%,^): +
43 + 56 = 99

-----
(program exited with code: 0)

Press any key to continue . . . |
```

```
C:\WINDOWS\SYSTEM32\cmd X + v
Enter the first number : 12
Enter the second number : 43
Enter the operator (+,-,*,/,%,^): -
12 - 43 = -31

-----
(program exited with code: 0)
```

```
C:\WINDOWS\SYSTEM32\cmd X + v
Enter the first number : 34
Enter the second number : 4
Enter the operator (+,-,*,/,%,^): /
34 / 4 = 8.5

-----
(program exited with code: 0)

Press any key to continue . . . |
```

```
C:\WINDOWS\SYSTEM32\cmd X + v
Enter the first number : 32
Enter the second number : 4
Enter the operator (+,-,*,/,%,^): %
32 % 4 = 0

-----
(program exited with code: 0)

Press any key to continue . . . |
```

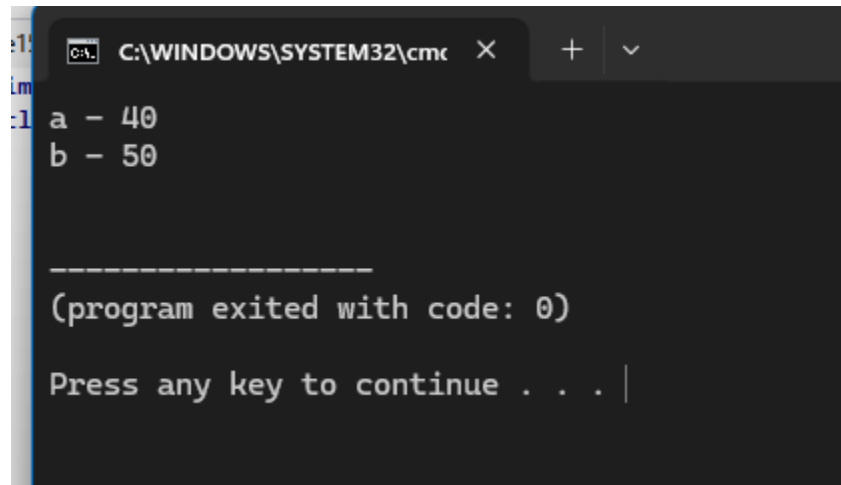
Q15.

```
import java.util.*;
class Example15{
    public static void swap(){
        int a = 50;
        int b = 40;

        a = a+b;
        b = a-b;
        a = a-b;

        System.out.println("a - "+a);
        System.out.println("b - "+b);
    }
    public static void main(String args[]){
        swap();
    }
}
```

Output.



```
C:\WINDOWS\SYSTEM32\cmd
a - 40
b - 50

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q16.

```
Example16.java X Example17.java X Example18.java X Example19.java X
1 import java.util.*;
2 class Example16{
3     public static void findP(){
4         Scanner sc = new Scanner(System.in);
5         System.out.print("Enater a number : ");
6         int num = sc.nextInt();
7         if(num<=0){
8
9         }else{
10            int tot = 0;
11            for(int i = 1;i<num;i++){
12                if(num%i==0){
13                    tot+=i;
14                }
15            }
16            if(tot==num){
17                System.out.print(num+" is a perfect number ");
18            }else{
19                System.out.print(num+" is not a perfect number ");
20            }
21        }
22    }
23
24    public static void main(String args[]){
25        findP();
26    }
27 }
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd X + v
Enater a number : 34
34 is not a perfect number

-----
(program exited with code: 0)
Press any key to continue . . . |
```

```
Enater a number : 6
6 is a perfect number

-----
(program exited with code: 0)
```

Q17.

```
Example17.java X Example18.java X Example19.java X
1  import java.util.*;
2  class Example17{
3      public static void findHMM(){ //HMM mean how many multi.
4          Scanner sc = new Scanner(System.in);
5          System.out.print("Enter the number that you want to get how many multiples in a range : ");
6          int num = sc.nextInt();
7
8          System.out.print("The lower bound of the range : ");
9          int r1 = sc.nextInt();
10
11         System.out.print("The upper bound of the range : ");
12         int r2 = sc.nextInt();
13         int n = 0;
14         for(int i = r1+1;i<r2;i++){
15             if(i%num==0){
16                 n++;
17             }
18         }
19         System.out.print("Multiples of "+num+" between "+r1+" and "+r2+" - "+n);
20     }
21     public static void main(String args[]){
22         findHMM();
23     }
24 }
25
26
```

Output.

```
Enter the number that you want to get how many multiples in a range : 3
The lower bound of the range : 10
The upper bound of the range : 100
Multiples of 3 between 10 and 100 - 30

-----
(program exited with code: 0)
```

Q18.

```
class Demo{
    public static void test(){
        System.out.println("Hello...");
    }
    public static void test(String name){
        System.out.println("Hello "+name+"...");
    }
    public static void main(String args[]){
        String s="java"; //Line 1
        test; //Line 2
        test(); //Line 3
        test(){ } //Line 4
        test(){ }; //Line 5
        Demo.test(); //Line 6
        System.out.println("test()"); //Line 7
        System.out.println(test()); //Line 8
        test(s); //Line 9
        System.out.println(test(s)); //Line 10
        System.out.println("test(s)"); //Line 11
    }
}
```

Answers:

line 1 – legal

* Standard String variable declaration and initialization.

line 2 – illegal

* Parentheses () are required when invoking/calling a method.

line 3 – legal

* Correct syntax for calling a static method with no arguments.

line 4 – illegal

* You cannot declare or define a method inside another method block (such as main).

line 5 – illegal

* Same as Line 4; method definitions are not allowed inside main.

line 6 – legal

* Valid static method invocation syntax (assuming a Demo class with a static test() method exists in the same package).

line 7 – legal

* Treats "test()" as a plain String literal inside quotes and prints it directly.

line 8 – illegal

* test() is a void method, meaning it returns nothing. You cannot pass a void method call as an argument to System.out.println().

line 9 – legal

* Correct syntax for calling the overloaded method test(String name) passing s as an argument.

line 10 – illegal

* test(s) is also a void method and has no return value to pass into System.out.println().

line 11 – legal

* Treats "test(s)" as a plain String literal inside quotes and prints it directly.

Q19.

```
ample19.java  Example20.java  Example21.java  Example22.java  Example23.java  Example24.java  Example25.java
import java.util.*;
class Example19{
    public static void getSub(){
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the Student name : ");
        String studentName = sc.nextLine();

        Random r =new Random();

        String [] g1 = {"Business & Accounting","Geography",
            "Citizenship Education","Entrepreneurship studies",
            "2nd language Sinhala","2nd Language Tamil","Foreign Languages(Arabic)",
            "Foreign Languages(Hindi)","Foreign Languages(French)","Foreign Languages(Japan)"};
        String [] g2 = {"Art","Tamil Literature","English Literature","Sinhala Literature","Music","Dancing"};
        String [] g3 = {"Information & technology","Agriculture","Home Economics",
            "Health Science","Art & Craft","Media"};

        String gro1 = g1[r.nextInt(g1.length)];
        String gro2 = g2[r.nextInt(g2.length)];
        String gro3 = g3[r.nextInt(g3.length)];

        System.out.println("Student Name: " + studentName);
        System.out.println("Group 01 Subject: " + gro1);
        System.out.println("Group 02 Subject: " + gro2);
        System.out.println("Group 03 Subject: " + gro3);
    }
    public static void main(String args[]){
        getSub();
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd  X  +  v
Enter the Student name : daham
Student Name: daham
Group 01 Subject: Entrepreneurship studies
Group 02 Subject: Tamil Literature
Group 03 Subject: Art & Craft

-----
(program exited with code: 0)
Press any key to continue . . . |
```

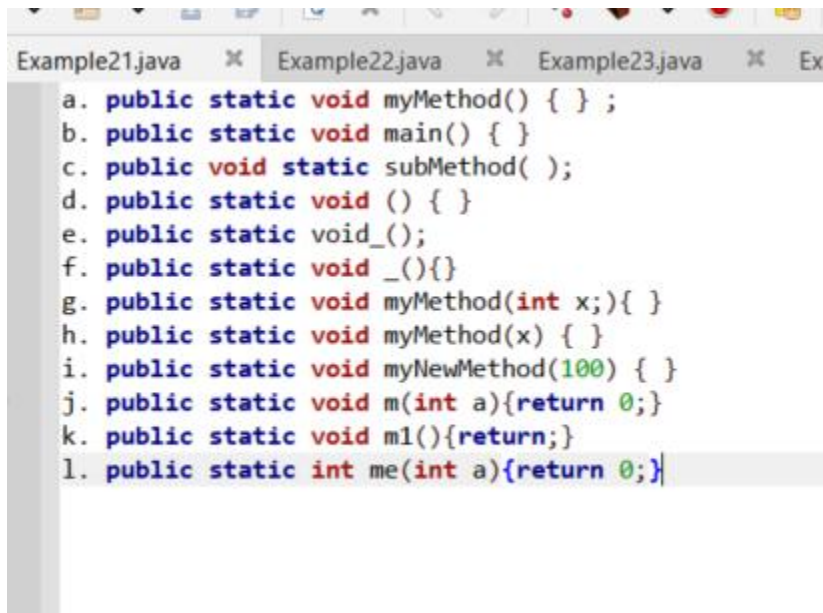
Q20.

```
Example20.java  Example21.java  Example22.java  Example23.java
1 class Test{
2     public static void testMethod(){
3         System.out.println("testMethod of Test");
4     }
5 }
6 class Demo{
7     public static void testMethod(){
8         System.out.println("testMethod of Demo");
9     }
10    public void testMethod2(){
11        System.out.println("testMethod of Demo");
12    }
13    public static void main(String args[]){
14        testMethod(); //line 1
15        testMethod2(); //line 2
16        Test.testMethod(); //line 3
17        Demo.testMethod(); //line 4
18        Demo.testMethod2(); //line 5
19        Demo d=new Demo(); //line 6
20        d.testMethod(); //line 7
21        d.testMethod2(); //line 8
22    }
23 }
24 }
```

Answer.

Line 2 and Line 5 - compile error because testMethod2()/Demo.testMethod2 are non static, and static method can't be directly call non static methods.

Q21.



```
Example21.java  Example22.java  Example23.java  Ex
a. public static void myMethod() { } ;
b. public static void main() { }
c. public void static subMethod( );
d. public static void ( ) { }
e. public static void _();
f. public static void _(){}
g. public static void myMethod(int x;){ }
h. public static void myMethod(x) { }
i. public static void myNewMethod(100) { }
j. public static void m(int a){return 0;}
k. public static void m1(){return;}
l. public static int me(int a){return 0;}
```

Answers.

- a. Legal
- b. Legal
- c. Illegal
 - * method body { } is missing
- d. Illegal
 - * missing method identifier(name)
- e. Illegal
 - * method body { } is missing
- f. legal
- g. Illegal
 - * we can't use semicolons when passing parameters
- h. Illegal
 - * Missing data type for parameter(x)
- i. illegal
 - * Missing data type and variable name
- j. illegal
- k. illegal
 - * without condition it was illegal
- l. legal

Q22.

```
ample22.java ✕ Example23.java ✕ Example24.java ✕ Example25.java ✕ Example26.java ✕ Examp
import java.util.*;
class Example22{
    public static double msConverter(double km){
        double ms = ((km *1000)/3600);
        return ms;
    }
    public static void main(String args[]){
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the Kilor Meteers per hour : ");
        double km = sc.nextDouble();

        double ms = msConverter(km);
        System.out.printf("speed in ms : %.2f ms^-1",ms);
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd
Enter the Kilor Meteers per hour : 345
speed in ms : 95.83 ms^-1

-----
(program exited with code: 0)

Press any key to continue . . .
```

Q23.

```
Example23.java  Example24.java  Example25.java  Example26.java  Example27.java  Examp
import java.util.*;
class Example23{
    public static int vCount(String str){
        int vcou = 0;
        for(int i = 0; i < str.length(); i++){
            char c = str.toUpperCase().charAt(i);
            if(c == 'A' || c == 'E' || c == 'I' || c == 'O' || c == 'U'){
                vcou++;
            }
        }
        return vcou;
    }
    public static void main(String args[]){
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter String values : ");
        String str = sc.nextLine();

        int vcou = vCount(str);
        System.out.print("The vowel count of that Strings : "+vcou);
    }
}
```

Output.

```
C:\WINDOWS\system32\cmd
Enter String values : weer
The vowel count of that Strings : 2

-----
(program exited with code: 0)

Press any key to continue . . . |
```

Q24.

```
Example24.java  Example25.java  Example26.java  Example27.java  Example28.java
import java.util.*;
class Example24{
    public static double findVol(double r, double h){
        double v = 3.14*r*r*h*1000;
        return v;
    }
    public static void main(String args[]){
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the radius(m) of cylinder : ");
        double r = sc.nextDouble();

        System.out.print("Enter the height(m) of cylinder : ");
        double h = sc.nextDouble();

        double v = findVol(r,h);
        System.out.printf("volume of the cylinder : %.3f L", v);
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd  +  v
Enter the radius(m) of cylinder : 3
Enter the height(m) of cylinder : 5
volume of the cylinder : 141300.000 L

-----
(program exited with code: 0)

Press any key to continue . . . |
```

Q25.

```
Example25.java  Example26.java  Example27.java  Example28.java
1  import java.util.*;
2  class Example25{
3      public static double findSV(double l){
4          double r = l/2.0;
5          double v = (4.0/3.0)*3.14*Math.pow(r,3);
6
7          return v;
8      }
9
10     public static void main(String args[]){
11         Scanner sc = new Scanner(System.in);
12         System.out.print("Enter the length(m) of the cube : ");
13         double l = sc.nextDouble();
14
15         double v = findSV(l);
16         System.out.printf("Volume of the largest sphere: %.3f L",v);
17     }
18 }
19
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd  +  v
Enter the length(m) of the cube : 5
Volume of the largest sphere: 65.417 L

-----
(program exited with code: 0)

Press any key to continue . . . |
```

Q26.

```
ample26.java ✕ Example27.java ✕ Example28.java ✕
class Example26{
    public static int increment(int x){
        x++;
        System.out.println("x : "+x);
        return x;
    }
    public static void main(String args[]){
        int x=100;
        System.out.println("x : "+x);
        increment(x);
        System.out.println("x : "+x);
        x=increment(x);
        System.out.println("x : "+x);
    }
}
```

Output.

```
x : 100
x : 101
x : 100
x : 101
x : 101
```

Explain .

System.out.println("x: " + x); (Line 1)

- In the main method, x is initialized to 100.
- Output: x: 100

increment(x); (Line 2)

- The value of x (100) is passed into the increment method.

- Inside increment, x++ increases the local variable x to 101.
- System.out.println("x: " + x); inside the method executes.
- Output: x: 101

System.out.println("x: " + x); (Line 3)

- In Java, primitive data types are passed by value. Changes made to x inside the increment method do not affect the original variable x in the main method.
- Therefore, x in main remains 100.
- Output: x: 100

x = increment(x); (Line 4)

- The increment(x) method is called again, printing x: 101 from inside the method. (Output: x: 101)
- This time, the returned value (101) is reassigned back to x in the main method (x = ...).

System.out.println("x: " + x); (Line 5)

- Now x in main has been updated to 101.
- Output: x: 101

Q27.

```
Example27.java  Example28.java
1  import java.util.*;
2  class Example27{
3      public static boolean findEIE(int num) {
4          num = Math.abs(num);
5          if (num == 0) {
6              return true;
7          }
8
9          for (; num > 0; num /= 10) {
10             int digit = num % 10;
11
12             if (digit % 2 != 0) {
13                 return false;
14             }
15
16             return true;
17
18         }
19
20     }
21
22     public static void main(String args[]){
23         Scanner sc = new Scanner(System.in);
24         System.out.print("Enter a integer value : ");
25         int number = sc.nextInt();
26
27         boolean result = findEIE(number);
28         System.out.println("Are all digits even? " + result);
29     }
30 }
31
32
33
```

Output.

```
Enter a integer value : 3445
Are all digits even? false
```

```
Enter a integer value : 2244
Are all digits even? true
```

Q28.

```
mple27.java ✕ Example28.java ✕
class Example{
    public static boolean isPass(double avg){
        //Insert code here //Line 10
    }
}
A. return;
B. return true;
C. return avg>=50;
D. if(avg>=50){return true;}else{return false;}
E. if(avg>=50){return true;}
F. return avg>=50 ? true:false;
G. if(avg>=50){return true;} return false;
```

Answers

B,C,D,G

Q29.

```
import java.util.*;
class Example29 {
    public static int getMDay(String x) {
        switch (x.toLowerCase()) {
            case "january":
            case "march":
            case "may":
            case "july":
            case "august":
            case "october":
            case "december":
                return 31;

            case "april":
            case "june":
            case "september":
            case "november":
                return 30;

            case "february":
                return 28;

            default:
                System.out.println("Month name incorrect !");
                return 0;
        }
    }

    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter Month Name : ");
        String month = input.nextLine();
        int days = getMDay(month);
        if (days != 0) {
            System.out.println("Number of days : " + days);
        }
        input.close();
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd.exe
Enter Month Name : march
Number of days : 31

-----
(program exited with code: 0)
Press any key to continue . . .
```

Q30.

```
Example37.java  Example36.java  Example35.java  Example34.java  Example32.java  Exa
import java.util.*;
class Example30 {
    public static void pOAndEDigits(int number) {
        String oddDigits = "";
        String evenDigits = "";

        int reversed = 0;
        int temp = number;

        if (temp == 0) {
            evenDigits = "0";
        }
        while (temp > 0) {
            int remainder = temp % 10;
            reversed = (reversed * 10) + remainder;
            temp /= 10;
        }
        while (reversed > 0) {
            int digit = reversed % 10;

            if (digit % 2 != 0) {
                oddDigits += digit;
            } else {
                evenDigits += digit;
            }
            reversed /= 10;
        }
        System.out.println(oddDigits + " " + evenDigits);
    }
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter positive integer: ");
        int num = input.nextInt();

        pOAndEDigits(num);
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd  +  v
Enter positive integer: 2345
35 24

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q31.

```
Example30.java  Example37.java  Example38.java  Example39.java  Example35.java
import java.util.*;
class Example31 {
    public static void pRolled(){
        Random r =new Random();
        int num1=r.nextInt(6)+1;
        int num2=r.nextInt(6)+1;
        int count = 0;
        while(num1!=num2){
            num1 = r.nextInt(6)+1;
            num2 = r.nextInt(6)+1;
            System.out.println(num1+" "+num2);
            count++;
        }
        System.out.println("Dice roll - "+count+" times");
    }
    public static void main(String[] args){
        pRolled();
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd  ×  +  v
6 5
4 6
6 3
5 2
2 3
1 1
Dice roll - 6 times

-----
(program exited with code: 0)
Press any key to continue . . .
```

Q32.

Line 1 can inserted A,B,C,G

and line2 can inserted A,B,C,E,G

Q33.

Answers

C,D,E,F,H,K,O

Q34.

```
import java.util.*;
class Example34 {
    public static long facto(int num) {
        long fact = 1;
        for (int i = 1; i <= num; i++) {
            fact *= i;
        }
        return fact;
    }
    public static long calGroups(int n, int r) {
        if (r > n || n < 0 || r < 0) {
            return 0;
        }
        long nFactorial = facto(n);
        long rFactorial = facto(r);
        long nMinusRFactorial = facto(n - r);

        return nFactorial / (nMinusRFactorial * rFactorial);
    }
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Number of children - ");
        int n = input.nextInt();

        System.out.print("Number of group members - ");
        int r = input.nextInt();

        long result = calGroups(n, r);
        System.out.println("The number of groups = " + result);
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd.exe
Number of children - 8
Number of group members - 4
The number of groups = 70

-----
(program exited with code: 0)

Press any key to continue . . .
```

Q35.

```
Example41.java  Example40.java  Example39.java  Example38.java  Example37.java  Example36.java  Example35.java
class Example35 {
    public static long findSM() {
        long number = 1;
        while (true) {
            boolean isDivisible = true;
            for (int i = 1; i <= 20; i++) {
                if (number % i != 0) {
                    isDivisible = false;
                    break;
                }
            }
            if (isDivisible) {
                return number;
            }
            number++;
        }
    }
    public static void main(String[] args) {
        long result = findSM();
        System.out.println("Smallest positive divisible by all of the number between 1 and 20: " + result);
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd  x  +  v
Smallest positive divisible by all of the number between 1 and 20: 232792560

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q36.

```
import java.util.*;
class Q36 {
    public static void checkQuadrant(double x, double y) {
        if (x < -5 || x > 5 || y < -5 || y > 5) {
            System.out.println("Error: Coordinates must be between -5 and 5 as shown in the grid.");
            return;
        }
        if (x > 0 && y > 0) {
            System.out.println("Quadrant I");
        } else if (x < 0 && y > 0) {
            System.out.println("Quadrant II");
        } else if (x < 0 && y < 0) {
            System.out.println("Quadrant III");
        } else if (x > 0 && y < 0) {
            System.out.println("Quadrant IV");
        } else {
            System.out.println("Origin");
        }
        double sumOfSquares = (x * x) + (y * y);
        System.out.println("Distance: " + sumOfSquares);
    }
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter x coordinate between (-5 to 5): ");
        double x = input.nextDouble();
        System.out.print("Enter y coordinate (-5 to 5): ");
        double y = input.nextDouble();

        checkQuadrant(x, y);
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd.exe
Enter x coordinate between (-5 to 5): 3
Enter y coordinate (-5 to 5): 4
Quadrant I
Distance: 25.0

-----
(program exited with code: 0)

Press any key to continue . . . |
```

Q37.

```
Example43.java  Example42.java  Example41.java  Example40.java  Example39.java
import java.util.*;
class Example37 {
    public static void cTriangle(double x, double y, double z) {
        if ((x + y > z) && (x + z > y) && (y + z > x)) {
            double perimeter = x + y + z;
            System.out.println("Valid triangle. Perimeter: " + perimeter);
        } else {
            System.out.println("Invalid triangle");
        }
    }
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter side x: ");
        double x = input.nextDouble();

        System.out.print("Enter side y: ");
        double y = input.nextDouble();

        System.out.print("Enter side z: ");
        double z = input.nextDouble();

        cTriangle(x, y, z);
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd  +  v
Enter side x: 5
Enter side y: 4
Enter side z: 8
Valid triangle. Perimeter: 17.0

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q38.

```
Example44.java  Example43.java  Example42.java  Example41.java  Example40.java  Example39.java  Example
import java.util.*;
class Example38 {
    public static void findC(double xA, double yA, double xB, double yB, double lambda, double mu) {
        double xC = (lambda * xB + mu * xA) / (lambda + mu);
        double yC = (lambda * yB + mu * yA) / (lambda + mu);

        System.out.println("Coordinates of C:");
        System.out.println("Xc: " + xC);
        System.out.println("Yc: " + yC);
    }
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter XA: ");
        double xA = input.nextDouble();

        System.out.print("Enter YA: ");
        double yA = input.nextDouble();

        System.out.print("Enter XB: ");
        double xB = input.nextDouble();

        System.out.print("Enter YB: ");
        double yB = input.nextDouble();

        System.out.print("Enter lambda (λ): ");
        double lambda = input.nextDouble();

        System.out.print("Enter mu (μ): ");
        double mu = input.nextDouble();

        findC(xA, yA, xB, yB, lambda, mu);
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd.exe
P
Enter XA: 234
Enter YA: 56
Enter XB: 34
Enter YB: 56
Enter lambda (?»): 56
Enter mu (?%): 67
Coordinates of C:
Xc: 142.9430894308943
Yc: 56.0

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q39.

```
Example45.java  Example44.java  Example43.java  Example42.java
class Example39 {
    public static void add(int a, int b) {
        System.out.println("Sum of integers: " + (a + b));
    }
    public static void add(double a, double b) {
        System.out.println("Sum of doubles: " + (a + b));
    }
    public static void add(String a, String b) {
        System.out.println("Concatenated String: " + (a + b));
    }
    public static void main(String[] args) {
        add(10, 20);
        add(5.5, 4.2);
        add("Hello, ", "World!");
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd  +  v
Sum of integers: 30
Sum of doubles: 9.7
Concatenated String: Hello, World!

-----
(program exited with code: 0)

Press any key to continue . . .
```

Q40.

```
1 import java.util.*;
2 class Example40 {
3     public static int absoluteDifference(int a, int b) {
4         int diff = a - b;
5         if (diff < 0) {
6             return -diff;
7         }
8         return diff;
9     }
10    public static double absoluteDifference(double a, double b) {
11        double diff = a - b;
12        if (diff < 0) {
13            return -diff;
14        }
15        return diff;
16    }
17    public static void main(String[] args) {
18        Scanner input = new Scanner(System.in);
19        System.out.print("Enter first integer: ");
20        int num1 = input.nextInt();
21        System.out.print("Enter second integer: ");
22        int num2 = input.nextInt();
23        int intResult = absoluteDifference(num1, num2);
24        System.out.println("Absolute difference of integers: " + intResult);
25        System.out.print("Enter first double: ");
26        double d1 = input.nextDouble();
27        System.out.print("Enter second double: ");
28        double d2 = input.nextDouble();
29        double doubleResult = absoluteDifference(d1, d2);
30        System.out.println("Absolute difference of doubles: " + doubleResult);
31    }
32 }
33 }
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd  X  +  v
Enter first integer: 56
Enter second integer: 5
Absolute difference of integers: 51
Enter first double: 4.56
Enter second double: 65.444
Absolute difference of doubles: 60.884

-----
(program exited with code: 0)
```

Q41.

```
import java.util.*;
class Example41 {
    public static double calculatePerimeter(double rad) {
        return 2 * 3.14 * rad;
    }
    public static double calculatePerimeter(double len, double wid) {
        if (len == wid) {
            return 4 * len;
        } else {
            return (2 * (len + wid));
        }
    }
    public static double calculatePerimeter(double sideA, double sideB, double sideC) {
        return sideA + sideB + sideC;
    }
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.println("--- circle ---");
        System.out.print("Enter radius of the circle : ");
        double circleRadius = sc.nextDouble();
        double circleAns = calculatePerimeter(circleRadius);
        System.out.println("Circle perimeter : " + circleAns);
        System.out.println();

        System.out.println("--- Square ---");
        System.out.print("Enter length : ");
        double squareLen = sc.nextDouble();
        System.out.print("Enter width : ");
        double squareWid = sc.nextDouble();
        double squareAns = calculatePerimeter(squareLen, squareWid);
        if (squareLen == squareWid) {
            System.out.println("Square perimeter : " + squareAns);
        } else {
            System.out.println("Error: Length and width must be equal for a Square!");
        }
        System.out.println();

        System.out.println("--- Rectangle ---");
        System.out.print("Enter length : ");
        double rectLen = sc.nextDouble();
        System.out.print("Enter width : ");
        double rectWid = sc.nextDouble();
        double rectAns = calculatePerimeter(rectLen, rectWid);
        if (rectLen != rectWid) {
            System.out.println("Rectangle perimeter : " + rectAns);
        } else {
            System.out.println("Error: Length and width cannot be equal for a Rectangle!");
        }
        System.out.println();

        System.out.println("--- Triangle ---");
        System.out.print("Enter side 1 of the Triangle: ");
        double firstSide = sc.nextDouble();
        System.out.print("Enter side 2 of the Triangle: ");
        double secondSide = sc.nextDouble();
        System.out.print("Enter side 3 of the Triangle: ");
        double thirdSide = sc.nextDouble();
        double triangleAns = calculatePerimeter(firstSide, secondSide, thirdSide);
        System.out.println("Triangle perimeter : " + triangleAns);
    }
}
```

Output.

```
CA\WINDOWS\SYSTEM32\cmd X + v
--- circle ---
Enter radius of the circle : 3
Circle perimeter : 18.84

--- Square ---
Enter length : 3
Enter width : 7
Error: Length and width must be equal for a Square!

--- Rectangle ---
Enter length : 7
Enter width : 4
Rectangle perimeter : 22.0

--- Triangle ---
Enter side 1 of the Triangle: 7
Enter side 2 of the Triangle: 6
Enter side 3 of the Triangle: 4
Triangle perimeter : 17.0

-----
(program exited with code: 0)
Press any key to continue . . . |
```

```
--- circle ---
Enter radius of the circle : 2
Circle perimeter : 12.56

--- Square ---
Enter length : 3
Enter width : 3
Square perimeter : 12.0

--- Rectangle ---
Enter length : 4
Enter width : 5
Rectangle perimeter : 18.0

--- Triangle ---
Enter side 1 of the Triangle: 6
Enter side 2 of the Triangle: 2
Enter side 3 of the Triangle: 5
Triangle perimeter : 13.0

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q42.

```
import java.util.*;
class Example42{
    public static boolean isPalindrome(int val) {
        int original = val;
        int flipped = 0;
        while (val > 0) {
            int rem = val % 10;
            flipped = (flipped * 10) + rem;
            val = val / 10;
        }
        return original == flipped;
    }
    public static boolean isPalindrome(String txt) {
        String reversed = "";
        for (int i = txt.length() - 1; i >= 0; i--) {
            reversed = reversed + txt.charAt(i);
        }
        return txt.equals(reversed);
    }
    public static void main(String[] args) {
        Scanner scan = new Scanner(System.in);

        System.out.print("Enter an integer number: ");
        int num = scan.nextInt();
        System.out.println("Is number palindrome? : " + isPalindrome(num));

        System.out.println();

        System.out.print("Enter a word: ");
        String str = scan.next();
        System.out.println("Is word palindrome? : " + isPalindrome(str));
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd
Enter an integer number: 234
Is number palindrome? : false

Enter a word: good
Is word palindrome? : false

-----
(program exited with code: 0)
```

```
C:\WINDOWS\SYSTEM32\cmd
Enter an integer number: 121
Is number palindrome? : true

Enter a word: wow
Is word palindrome? : true

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q43.

```
Example49.java  Example48.java  Example47.java  Example46.j
import java.util.*;
class Example43 {
    public static String convertToBinary(int val) {
        String bin = "";
        while (val > 0) {
            int rem = val % 2;
            bin = rem + bin;
            val = val / 2;
        }
        if (bin.equals("")) {
            bin = "0";
        }
        return bin;
    }
    public static String convertToOctal(int val) {
        String octa = "";
        while (val > 0) {
            int rem = val % 8;
            octa = rem + octa;
            val = val / 8;
        }
        if (octa.equals("")) {
            octa = "0";
        }
        return octa;
    }
    public static String convertToHexadecimal(int val) {
        String hexa = "";
        while (val > 0) {
            int rem = val % 16;

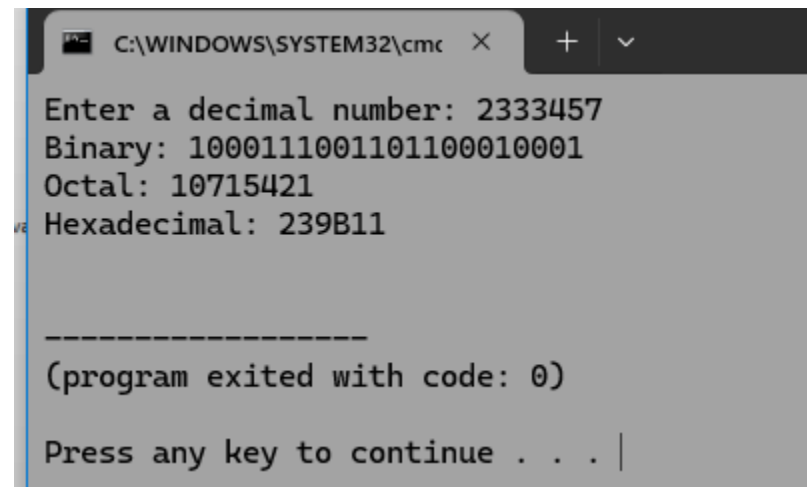
            if (rem == 10) {
                hexa = "A" + hexa;
            } else if (rem == 11) {
                hexa = "B" + hexa;
            } else if (rem == 12) {
                hexa = "C" + hexa;
            } else if (rem == 13) {
                hexa = "D" + hexa;
            } else if (rem == 14) {
                hexa = "E" + hexa;
            } else if (rem == 15) {
                hexa = "F" + hexa;
            } else {
                hexa = rem + hexa;
            }

            val = val / 16;
        }
        if (hexa.equals("")) {
            hexa = "0";
        }
        return hexa;
    }
    public static void main(String[] args) {
        Scanner scan = new Scanner(System.in);

        System.out.print("Enter a decimal number: ");
        int num = scan.nextInt();

        System.out.println("Binary: " + convertToBinary(num));
        System.out.println("Octal: " + convertToOctal(num));
        System.out.println("Hexadecimal: " + convertToHexadecimal(num));
    }
}
```

Output.



```
C:\WINDOWS\SYSTEM32\cmd X + v
Enter a decimal number: 2333457
Binary: 1000111001101100010001
Octal: 10715421
Hexadecimal: 239B11

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q44.

```
Example50.java  Example49.java  Example48.java  Example47.java  Example46.java
import java.util.*;
class Example44 {
    public static int findFactorial(int val) {
        int fact = 1;

        for (int i = 1; i <= val; i++) {
            fact = fact * i;
        }
        return fact;
    }
    public static void main(String[] args) {
        Scanner scan = new Scanner(System.in);

        System.out.print("Enter an integer number: ");
        int num = scan.nextInt();
        int fact = findFactorial(num);
        System.out.println("Factorial: " + fact);
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd  +  v
Enter an integer number: 6
Factorial: 720

-----
(program exited with code: 0)
Press any key to continue . . . |
```

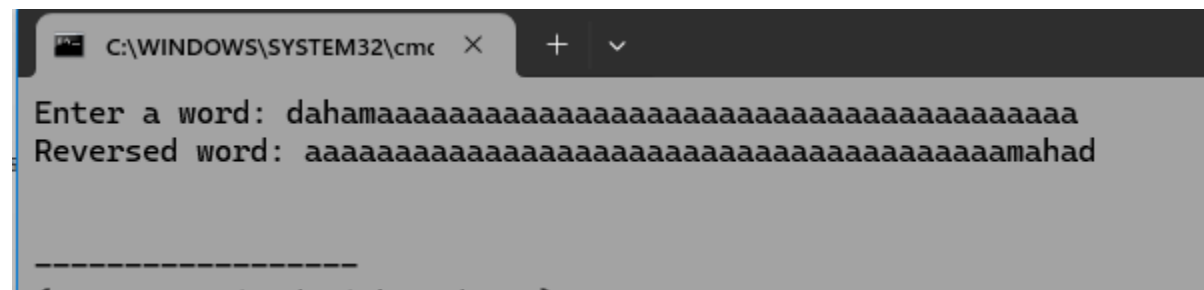
Q45.

```
import java.util.*;
class Example45 {
    public static String reverseString(String txt) {
        String reversed = "";

        for (int i = txt.length() - 1; i >= 0; i--) {
            reversed = reversed + txt.charAt(i);
        }
        return reversed;
    }
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter a word: ");
        String str = input.nextLine();
        String reversestr =reverseString(str);
        System.out.println("Reversed word: " +reversestr );
    }
}
```

Output.



```
C:\WINDOWS\SYSTEM32\cmd
Enter a word: daham
Reversed word: aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaamahad
```

Q46.

```
imple50.java  Example49.java  Example48.java  Example47.java  Example46.java
import java.util.*;
class Example46 {
    public static void getGeometricSeries(long r, long a) {
        long currentTerm = a;
        for (int i = 1; i <= 10; i++) {
            System.out.print(currentTerm + " ");
            currentTerm = currentTerm * r;
        }
    }
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter 1st term (r): ");
        long r = input.nextLong();

        System.out.print("Enter common ratio (a): ");
        long a = input.nextLong();

        System.out.println("First 10 terms of the geometric series:");

        getGeometricSeries(a, r);
    }
}
```

Output.

```
Enter common ratio (r): 6
Enter 1st term (a): 5
First 10 terms of the geometric series:
6 30 150 750 3750 18750 93750 468750 2343750 11718750

-----
(program exited with code: 0)

Press any key to continue . . . |
```

Q47.

```
Example47.java
import java.util.*;
class Example47 {
    public static void power(int a,int b){
        int power_0 = (int)Math.pow(a,b);
        System.out.println("Result : "+power_0);
    }
    public static void main(String[] args){
        Scanner input=new Scanner(System.in);

        System.out.print("Enter base number      : ");
        int num1 = input.nextInt();

        System.out.print("Enter exponent (power) : ");
        int num2 = input.nextInt();

        power(num1,num2);
    }
}
```

Output

```
system.out.print( Enter common ra
C:\WINDOWS\SYSTEM32\cmd
Enter base number      : 7
Enter exponent (power) : 8
Result : 5764801

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q48.

```
Example50.java  Example49.java  Example48.java
1  import java.util.*;
2  class Example48{
3      public static int sumOfDigits(int num){
4          int sum=0;
5          while(num!=0){
6              sum+=num%10;
7              num/=10;
8          }
9          return sum;
10     }
11     public static void main(String[] args){
12         Scanner input=new Scanner(System.in);
13         System.out.print("Input an integer : ");
14         int num=input.nextInt();
15         int sum=sumOfDigits(num);
16         System.out.println("Sum of digits for "+num+" : "+sum);
17     }
18 }
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd  +  v
Input an integer : 234567
Sum of digits for 234567 : 27

-----
(program exited with code: 0)

Press any key to continue . . . |
```

Q49.

```
import java.util.*;
class Example49 {
    public static void tryguessing(int r_num){
        Scanner input=new Scanner(System.in);
        for(int i=1;i<=5;i++){
            System.out.print("Enter your guessing number : ");
            int num0 = input.nextInt();
            if(num0==r_num){
                System.out.println("Your Guessing is correct.");
            }else{
                if(num0 > r_num){
                    System.out.println("Too high, try again.");
                }else{
                    System.out.println("Too low, try again.");
                }
            }
        }
        System.out.println("Maximum attempts exceeded ");
    }
    public static void main(String[] args){
        Scanner input=new Scanner(System.in);

        Random r = new Random();

        int r_num = r.nextInt(100)+1;

        tryguessing(r_num);
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd  +  v
Enter your guessing number : 56
Too high, try again.
Enter your guessing number : 34
Too low, try again.
Enter your guessing number : 45
Too low, try again.
Enter your guessing number : 50
Too low, try again.
Enter your guessing number : 53
Your Guessing is correct.
Maximum attempts exceeded

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q50.

```
import java.util.Scanner;

public class Me {
    static String password = "1234";
    static int attempts = 0;
    public static void displayDetails() {
        System.out.println("--- User Details ---");
        System.out.println("Name: Daham Wijemanna");
        System.out.println("Age: 22");
        System.out.println("Occupation: Software Developer");
    }
    public static void checkPassword() {
        Scanner sc = new Scanner(System.in);
        while (attempts < 3) {
            System.out.print("Enter password: ");
            String input = sc.nextLine();
            attempts++;
            if (input.equals(password)) {
                displayDetails();
                return;
            }
            if (attempts == 3) {
                System.out.println("Maximum attempts reached.");
                return;
            }
            String choice = "";
            while (!choice.equals("Yes") && !choice.equals("yes") && !choice.equals("Y") && !choice.equals("y") &&
                !choice.equals("No") && !choice.equals("no") && !choice.equals("N") && !choice.equals("n")) {

                System.out.print("Do you want to continue (Yes/No)? ");
                choice = sc.nextLine();

                if (!choice.equals("Yes") && !choice.equals("yes") && !choice.equals("Y") && !choice.equals("y") &&
                    !choice.equals("No") && !choice.equals("no") && !choice.equals("N") && !choice.equals("n")) {
                    System.out.println("Invalid input. Try again.");
                }
            }
            if (choice.equals("No") || choice.equals("no") || choice.equals("N") || choice.equals("n")) {
                System.out.println("Program ended.");
                return;
            }
        }
    }
    public static void main(String[] args) {
        checkPassword();
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd
Enter password: 123
Do you want to continue (Yes/No)? y
Enter password: 456
Do you want to continue (Yes/No)? h
Invalid input. Try again.
Do you want to continue (Yes/No)? y
Enter password: 1234
Name: Daham Wijemanna
Age: 22
Occupation: Software Developer

-----
(program exited with code: 0)
Press any key to continue . . . |
```

```
Enter password: 123
Do you want to continue (Yes/No)? y
Enter password: 434
Do you want to continue (Yes/No)? y
Enter password: 234
Maximum attempts reached.
```

```
-----
(program exited with code: 0)

Press any key to continue . . . |
```

```
C:\WINDOWS\SYSTEM32\cmd  X + v
Enter password: 234
Do you want to continue (Yes/No)? n
Program ended.
```

```
-----
(program exited with code: 0)

Press any key to continue . . . |
```

